

Section 12.2: Vectors

Vectors and the Geometry of Space

MTH 310
Calculus III

Scalars vs. Vectors

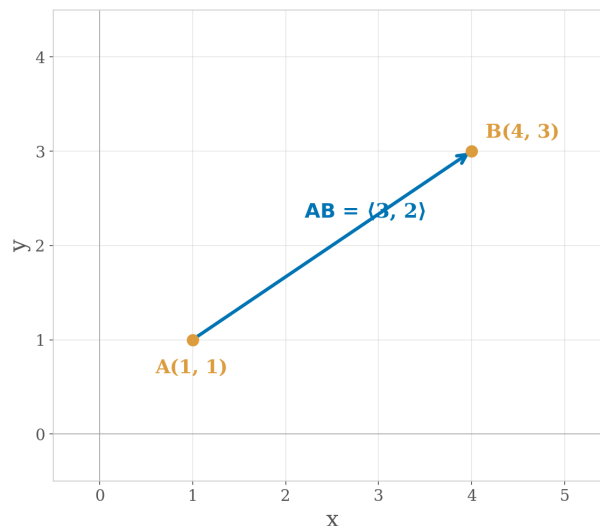
A **scalar** has magnitude only. A **vector** has both magnitude AND direction.

Displacement Vectors and Equal Vectors

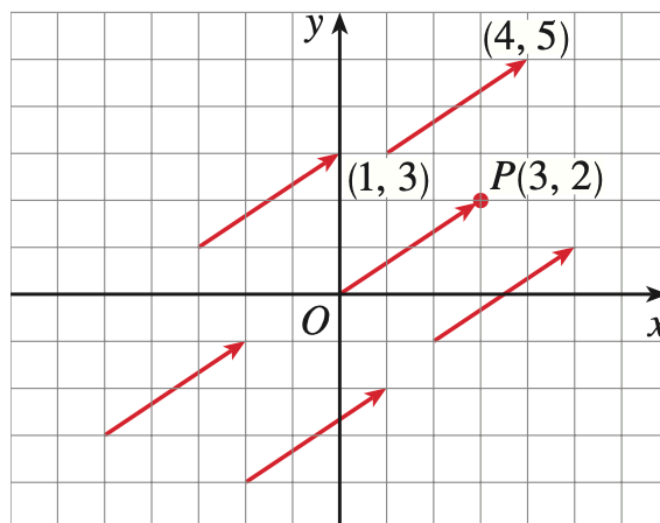
Equal Vectors

Two vectors are **equal** if they have the same magnitude and the same direction — even if they start at different points.

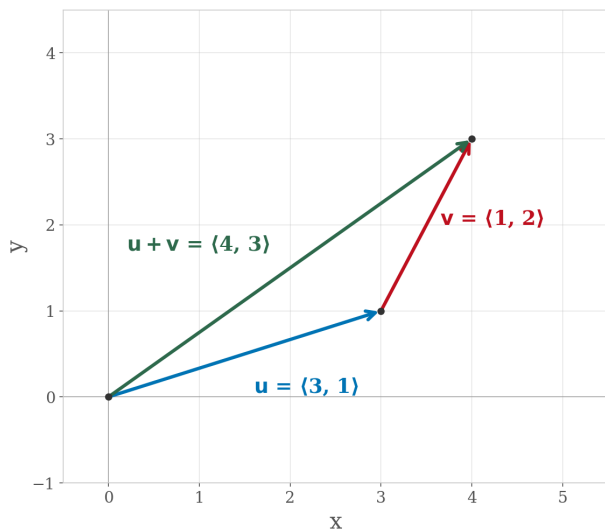
Equal vectors can be **translated** (slid) anywhere in space without changing the vector.



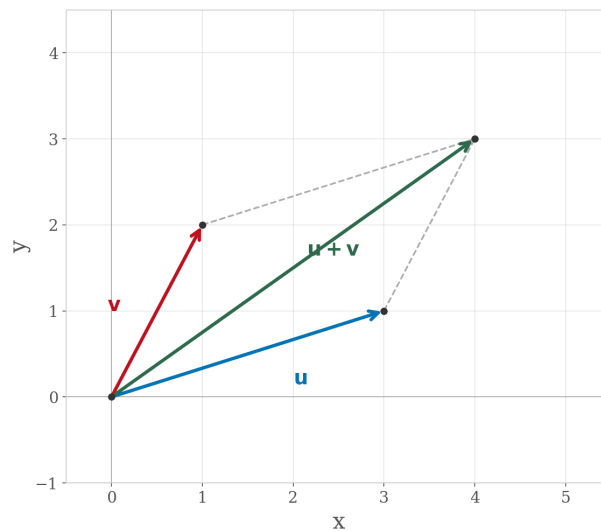
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Vector Addition



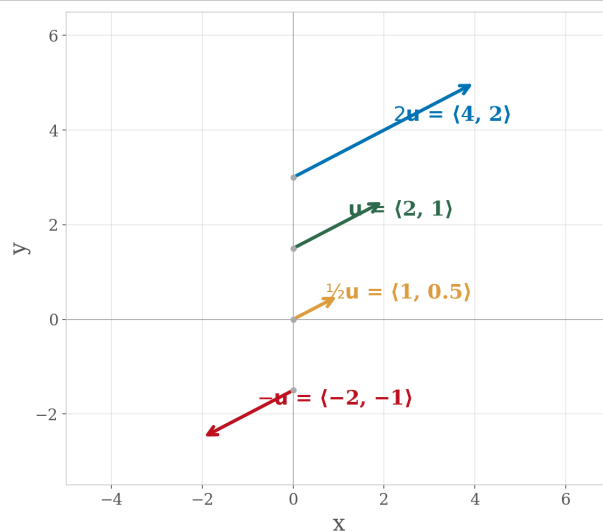
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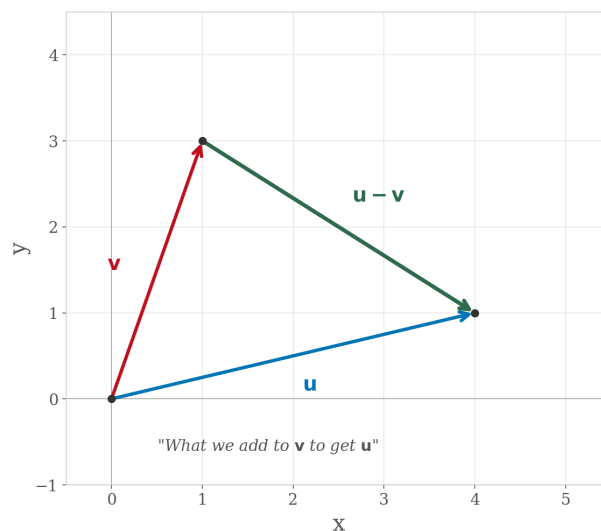
Vector addition is **commutative**: $\vec{u} + \vec{v} = \vec{v} + \vec{u}$. Both the triangle and parallelogram laws give the same result.

Scalar Multiplication and Vector Difference

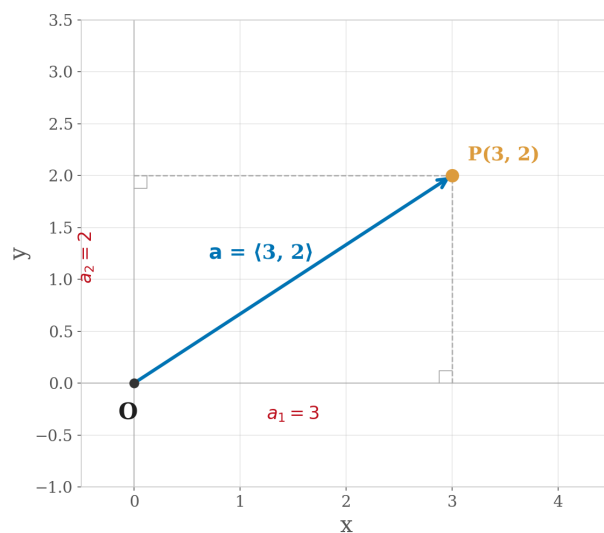


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Think of $\vec{u} - \vec{v}$ as "what you add to $\vec{v}$ to get $\vec{u}$ " — like how $10 - 7.36 = 2.64$ answers "what do I add to 7.36 to get 10?"



Components



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Displacement Vector Formula

Displacement Vector Formula

Given points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$, the displacement vector from A to B is:

$$\vec{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Terminal minus initial.

Example 1

Find the displacement vector $\vec{AB}$ for $A(2, -3, 4)$ and $B(-2, 1, 1)$.

Subtract: terminal minus initial, component by component.

Magnitude

Magnitude of a Vector

The **magnitude** (or length) of a vector $\vec{a} = \langle a_1, a_2, a_3 \rangle$ is:

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

In 2D: $|\vec{a}| = \sqrt{a_1^2 + a_2^2}$

Example: Vector Arithmetic

Example 2

For $\vec{a} = \langle 2, -1, 3 \rangle$ and $\vec{b} = \langle 0, 4, -3 \rangle$, compute:

1. $-3\vec{b}$
2. $|-3\vec{b}|$
3. $2\vec{a} + 5\vec{b}$

Compute each quantity step by step.

Your Turn: Vector Arithmetic Practice

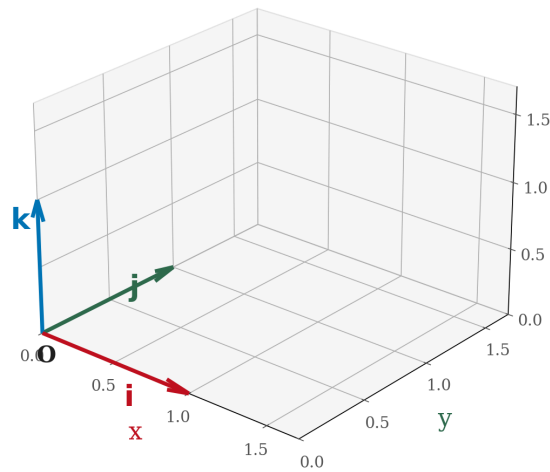
Example 3

Your turn! For $\vec{u} = \langle 3, -1, 2 \rangle$ and $\vec{v} = \langle -1, 4, 1 \rangle$, find:

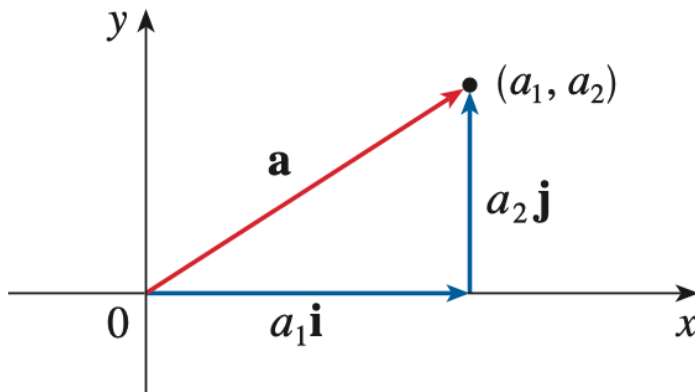
1. $2\vec{u} - 3\vec{v}$
2. $|2\vec{u} - 3\vec{v}|$

Work through the scalar multiplications first, then subtract, then find the magnitude.

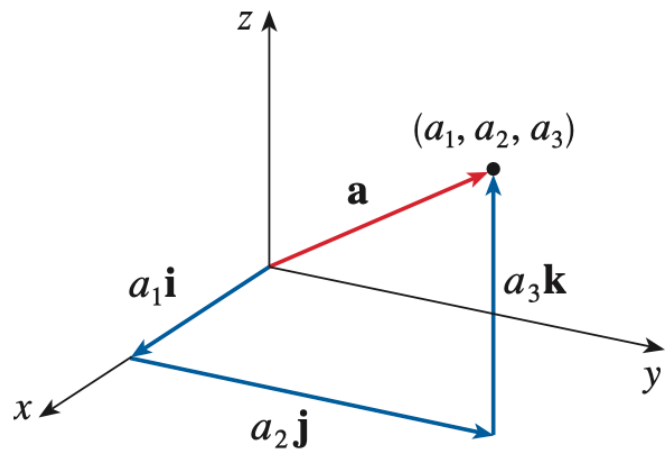
Standard Basis Vectors



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(a) $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$



(b) $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

Unit Vectors

Unit Vector

A **unit vector** is a vector with magnitude 1.

To find the unit vector in the direction of $\vec{a}$:

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

This process is called **normalizing** the vector.

Example 4

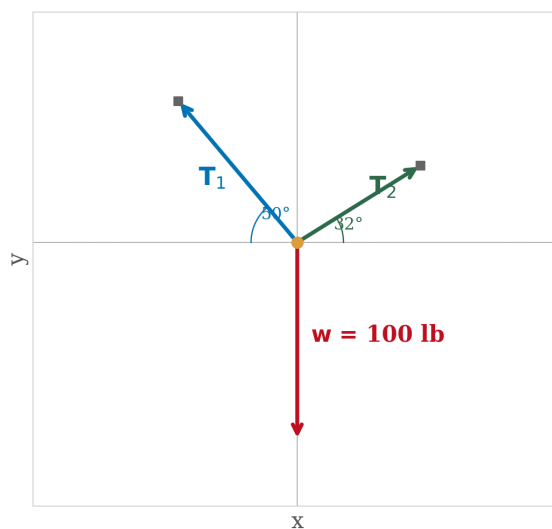
Find the unit vector in the direction of $\vec{a} = \langle 3, 4 \rangle$.

Find $|\mathbf{a}|$, then divide each component by it.

Application: Forces in Equilibrium

Example 5

A 100-lb weight hangs from two wires as shown. Find the tensions $\vec{T}_1$ and $\vec{T}_2$ in the wires and the magnitudes of these tensions.



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Homework

Section 12.2: 1, 2, 3, 4, 8, 9, 11, 13, 15, 17, 19, 21, 23, 25